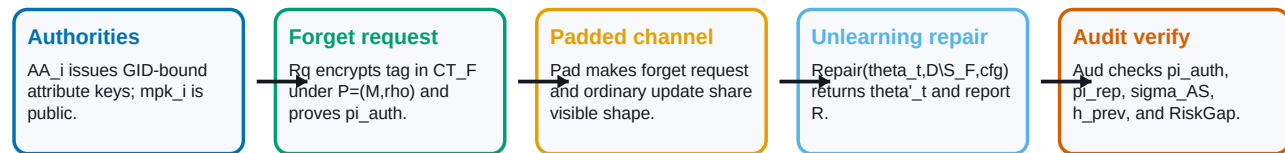


MA-ABE-FU protocol surface and UCAP evidence fields

Control plane authorization and learning-plane unlearning repair are separated, then transcript-bound for audit.



UCAP evidence object O_F

c_P=H(P)

policy commitment

H(ch)

padded-channel transcript hash

H(θ'_t)

post-repair model commitment

π_{auth}

ZK proof of authorized policy satisfaction

σ_{AS}

server signature over the evidence object

c_s=H(scope)

forget request scope commitment

H(θ_t)

pre-repair model commitment

H(R)

residual-risk report commitment

π_{rep}

ZK proof of unlearning repair consistency

h_{prev}

previous UCAP hash-chain pointer

The server receives commitments and proof handles, not plaintext policy attributes; model influence is measured by RiskGap.